

Detection of sarcocystis parasites in retail beef: a regional survey combining histological and genetic detection methods.



Sarcocystis spp. are parasitic protists acquired when undercooked, cyst-laden meat is consumed. While both *Sarcocystis hominis* and *S. cruzi* encyst in beef, only *S. hominis* is pathogenic to humans. In this study, we used histological methods and novel molecular techniques to determine the regional prevalence and identity of *Sarcocystis* spp. in retail beef. Of 110 samples, 60 supported amplification of parasite rRNA by PCR. All 41 sequenced representatives were identified as *S. cruzi*. To compare detection methods, 48 samples were then examined in parallel by histology and PCR, and 16 and 26 samples, respectively, were positive. Five samples positive by initial histologic sections were not amplified by PCR. Fifteen PCR-positive samples did not contain sarcocysts on initial histologic section, but additional sections from these samples revealed sarcocysts in an additional 12 samples. When combined, histology with additional sections and PCR detected 31 positive specimens of the 48 total specimens. We found no evidence of human pathogen *S. hominis* and confirm that cattle pathogen *S. cruzi* is highly prevalent in this regional sample. PCR assays may increase the detection sensitivity of *Sarcocystis* spp. and contribute diagnostic precision.